3 **(a)** Parallel light from two light sources of the same wavelength of 590 nm are incident on a slit. The angle between the two beams of light is θ, as shown in Fig. 3.1.

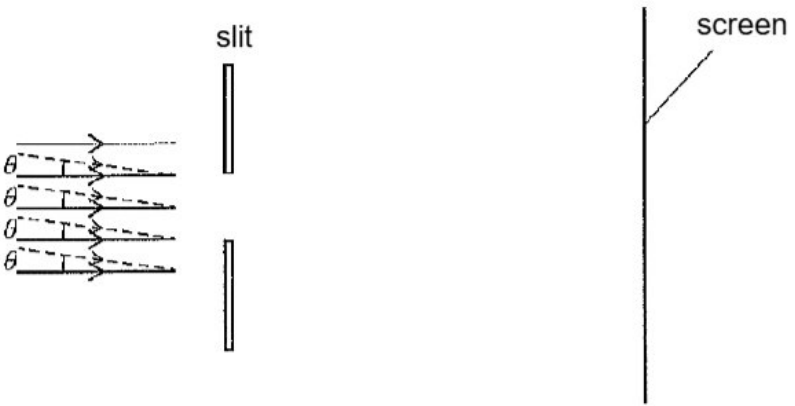


Fig. 3.1 (not drawn to scale)

Each beam forms a separate interference pattern on the screen.

(i) Explain how the single slit diffraction pattern produced by a beam can be formed on the screen.

.....

.....

.....

..... [2]

(ii) The angle between the two incident beams is gradually increased from zero.

Using Rayleigh criterion, describe and explain any change in the appearance of the two diffraction patterns.

.....

.....

.....

..... [2]

(iii) The distance between the screen and slit is 2.4 m and the slit width is 0.60 mm.

Determine the minimum angle θ for which the two interference patterns are just resolved.

θ = ... rad [2]

(b) A parallel beam of white light that consists of wavelengths from 350 nm to 650 nm is incident normally on a diffraction grating that has 500 lines per millimetre.

(i) Calculate the maximum angle of diffraction in the second-order spectrum.

θ = ... [2]

(ii) Hence, determine whether the second-order and third-order spectra overlap.

Show your working clearly.

..... [2]

(c) A scientist wishes to distinguish between two closely spaced wavelengths using a diffraction grating. The scientist may prefer to observe the second-order spectrum instead of the first-order spectrum.

State one advantage and disadvantage of this choice.

Advantage: .....

.....

Disadvantage: .....

..... [2]